

Data Warehousing

Lecture- 4: DW Modelling

Ali Zidane ElQutaany

Types of Facts/Measures

❖ Fully Additive Measures/Facts → Example

Numeric Features → {TransactionId, QuantitySold, UnitPrice, SalesAmount}

~~Text Features → DateTime, StoreName, ProductId~~

DateTime	StoreName	ProductId	TransactionId	QuantitySold	UnitPrice	SalesAmount
1/4/2022	Faisal	Prod-1	100	10	100	1000
1/4/2022	Faisal	Prod-1	200	20	100	2000
1/4/2022	Giza	Prod-2	300	30	200	6000
1/4/2022	Giza	Prod-3	400	20	200	4000
1/5/2022	Dokki	Prod-1	500	15	100	1500
1/5/2022	Haram	Prod-1	600	100	100	10000
1/5/2022	Haram	Prod-5	700	102	15	3030
1/6/2022	Asuit	Prod-4	800	12	100	1200
1/6/2022	Minya	Prod-2	1000	204	200	40800

Types of Facts/Measures

➤ The numeric measures in a fact table fall into three categories or types:

- ❖ **Fully Additive Measures/Facts:** Measures that can be summed across all dimensions.
- ❖ **Semi-Additive Measures/Facts:** Measures that can be summed across some dimensions but NOT all, **usually NOT across time.**
- ❖ **Non-Additive Measures / Facts:** Measures that cannot be summed across any of the dimensions.

➤ Examples:

- ❖ **Fully Additive Measures/Facts:**
 - ❖ Sales Amount
 - ❖ Quantity Sold
 - ❖ Revenue
 - ❖ Cost
 - ❖ Units Produced
 - ❖ Student Grades

Types of Facts/Measures

- ❖ **Semi-Additive Measures/Facts:** Measures that can be summed across some dimensions but NOT all, usually NOT across time.

➤ Examples:

- ❖ **Semi-Additive Measures/Facts:**

- ❖ Inventory Level
- ❖ Bank Balance
- ❖ Stock Quantity on Hand
- Inventory:
 - Jan 31 = 100 units
 - Feb 28 = 120 units
- You **cannot sum**: $100 + 120 = 220$ ❌ (wrong)
- Instead, you use:
 - Last value
 - Average
 - Min/Max

Types of Facts/Measures

- ❖ **Non-Additive Measures / Facts:** Measures that cannot be summed across any dimension.

➤ Examples:

- ❖ **Semi-Additive Measures/Facts:**

- ❖ Profit Margin %

- ❖ Discount %

- ❖ Ratio

- ❖ Conversion Rate

- If:

- Product A margin = 10%

- Product B margin = 20%

- You **cannot** sum:

- $10\% + 20\% = 30\%$ ❌ (incorrect)

- Instead, calculate:

- Weighted average

- ✓ Never summed directly.

Types of Facts/Measures

Type	Add Across Time?	Add Across Other Dimensions?	Example
Additive			Sales Amount
Semi-Additive			Inventory
Non-Additive			Profit Margin %

Types of Facts/Measures

- The numeric measures in a fact table fall into three categories or types:
 - ❖ ~~Fully Additive Measures/Facts.~~
 - ❖ Semi-Additive Measures/Facts.
 - ❖ Non-Additive Measures / Facts.
- Semi-additive measures can be aggregated across some dimensions, but not all.
- For example: Checking account or savings account balance amounts are common semi-additive facts.
- They are **additive** across all dimensions except **TIME**.
- It does not make any sense to **aggregate** the balance amounts from October, November, and December for the same account. **But can aggregate balance for different accounts on the same month**

Types of Facts/Measures

❖ Semi-Additive Measures/Facts.

DateTime	StoreName	ProductId	TransactionId	Quantity Sold	Unit Price	Sales Amount	QTYinHand	Tax
1/4/2022	Faisal	Prod-1	100	10	100	1000	100	12.00%
1/4/2022	Faisal	Prod-1	200	20	100	2000	80	12.00%
1/4/2022	Giza	Prod-2	300	30	200	6000	200	12.00%
1/4/2022	Giza	Prod-3	400	20	200	4000	300	12.00%
1/5/2022	Dokki	Prod-1	500	15	100	1500	300	12.00%
1/5/2022	Haram	Prod-1	600	100	100	10000	400	12.00%
1/5/2022	Haram	Prod-5	700	102	15	3030	100	12.00%
1/6/2022	Asuit	Prod-4	800	12	100	1200	100	12.00%
1/6/2022	Minya	Prod-2	1000	204	200	40800	100	12.00%

Types of Facts/Measures

❖ Semi-Additive Measures/Facts.

DateTime	StoreName	ProductId	TransactionId	Quantity Sold	Unit Price	Sales Amount	QTYinHand	Tax
1/4/2022	Faisal	Prod-1	100	10	100	1000	100	12.00%
1/4/2022	Faisal	Prod-1	200	20	100	2000	80	12.00%
1/4/2022	Giza	Prod-2	300	30	200	6000	200	12.00%
1/4/2022	Giza	Prod-3	400	20	200	4000	300	12.00%
1/5/2022	Dokki	Prod-1	500	15	100	1500	300	12.00%
1/5/2022	Haram	Prod-1	600	100	100	10000	400	12.00%
1/5/2022	Haram	Prod-5	700	102	15	3030	100	12.00%
1/6/2022	Asuit	Prod-4	800	12	100	1200	100	12.00%
1/6/2022	Minya	Prod-2	1000	204	200	40800	100	12.00%

Types of Facts/Measures

- The numeric measures in a fact table fall into three categories or types:
 - ❖ ~~Fully Additive Measures/Facts.~~
 - ❖ ~~Semi-Additive Measures/Facts.~~
 - ❖ Non-Additive Measures / Facts.
- Finally, some measures are completely non-additive, such as ratios, GPAs, ...
- As a conclusion, whenever the aggregation of a measure is useful ONLY when it is done per ZERO dimensions, here it is non-additive measure!!

Types of Facts/Measures

- The numeric measures in a fact table fall into three categories or types:
 - ❖ ~~Fully Additive Measures/Facts.~~
 - ❖ ~~Semi-Additive Measures/Facts.~~
 - ❖ Non-Additive Measures / Facts.
- Emphasizing that, usually, non-additives are averages, percentages and ratios and it is irrelevant to perform additional aggregations on them.
- For example, summing up 2 selling ratio of a product will not be meaningful at all.
- Thus, non-additive facts should not be stored within the fact table and will usually be manipulation on the additive facts within the fact table.

Types of Fact Tables

- **Three Types of Fact Tables:**

- ❖ Transaction Fact Table
- ❖ Factless Fact Table.
- ❖ Periodic Snapshot Fact Tables.
- ❖ Cumulative Fact Tables.

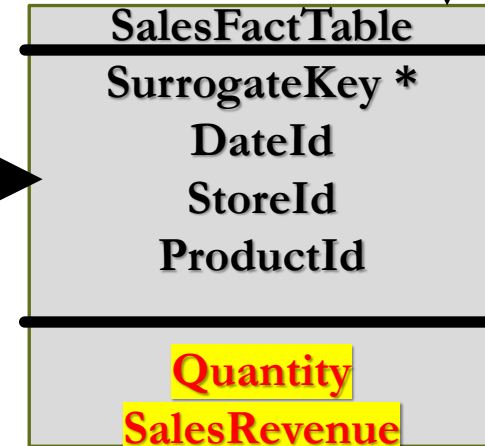
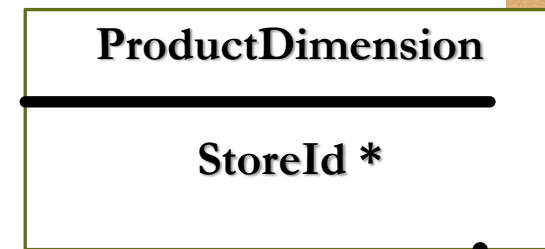
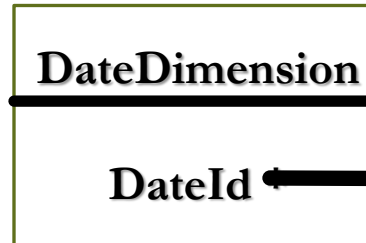
- **Granularity/Grain:**

- ❖ The lowest level of detail for a particular business purpose or for an attribute is called the grain or granularity.
- ❖ For example, if we consider the date/time dimension, it could start at the year (topmost), month, quarter, period, week, day, hour, minute, second (lowest) level of granularity.
- ❖ Most data warehouses do not go to the seconds or hundredths of seconds level, but it could be possible based on the business processes and the requirements.
- ❖ The same for Country, State, City, ...
- ❖ The same for Category, Brand, Product,
- ❖ The same for University, Faculty, Department, ...

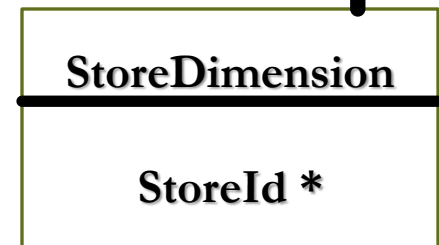
Types of Fact Tables

Three Types of Fact Tables:

- ❖ **Transaction Fact Table**
- ❖ **Factless Fact Table.**
 - ✓ Periodic Snapshot Fact Tables.
 - ✓ Cumulative Snapshot Periodic Fact Tables:



DateId	StoreId	ProductId	Quantity	SalesRevenue
1/4/2022	Faisal	Prod-1	10	1000
1/4/2022	Faisal	Prod-1	20	2000
1/5/2022	Haram	Prod-5	102	3030
1/6/2022	Asuit	Prod-4	12	1200
1/6/2022	Minya	Prod-2	204	40800



Types of Fact Tables

Transaction Fact Table:

- ✓ Simple Structure.
- ✓ Keeps most detailed level (Lowest Grain of data)
- ✓ Easy to aggregate.
- ✓ Mostly ADDITIVE Facts.

Types of Fact Tables

➤ Three types of Fact Tables:

❖ ~~Transactions Fact Table (Our Popular Fact Table) to save the facts per the minimum grain of time. It contains ONLY ADDITIVE MEASURES.~~

❖ Factless Fact Table:

- It is a fact table with no measures → associate entity between n tables with binary n-1 ary relationship.
- In the real world, it is possible to have a fact table that contains no measures or facts. These tables are called “Factless Fact tables”.
- E.g.: A fact table which has only product key and date key is a Factless fact. There are no measures in this table. But still you can get the number products sold over a period.
 - ✓ Periodic Snapshot Fact Tables.
 - ✓ Cumulative Snapshot Periodic Fact Tables:

Types of Fact Tables

❖ Periodic Snapshot Fact Tables.

- Periodic snapshot tables record the cumulative performance of the business at predefined periods of time.
- A predetermined interval for taking snapshots is the key: daily, weekly, monthly, etc.
- A fact table that stores data at regular intervals (daily, weekly, monthly), regardless of whether activity occurred.
- The results are saved in the **Periodic Snapshot Fact Table**.
- Characteristics:
 - One row per period (e.g., per day per product).
 - Includes summary data for that time slice.
 - Dense table (rows exist even if activity = 0).
 - Example: Each day has a record.

Date	Product	Inventory Qty	Daily Sales
Feb 1	A	100	20
Feb 2	A	120	15
Feb 3	A	110	18

Types of Fact Tables

❖ Periodic Snapshot Fact Tables.

- Business use cases:
 - Daily sales summary
 - Daily inventory snapshot
 - Weekly performance reports
 - Bank daily balances
 - Weekly ...
 - Monthly ...
 - Quarterly
 - Half Yearly
 - Yearly

Types of Fact Tables

❖ **Periodic Snapshot Fact Tables.**

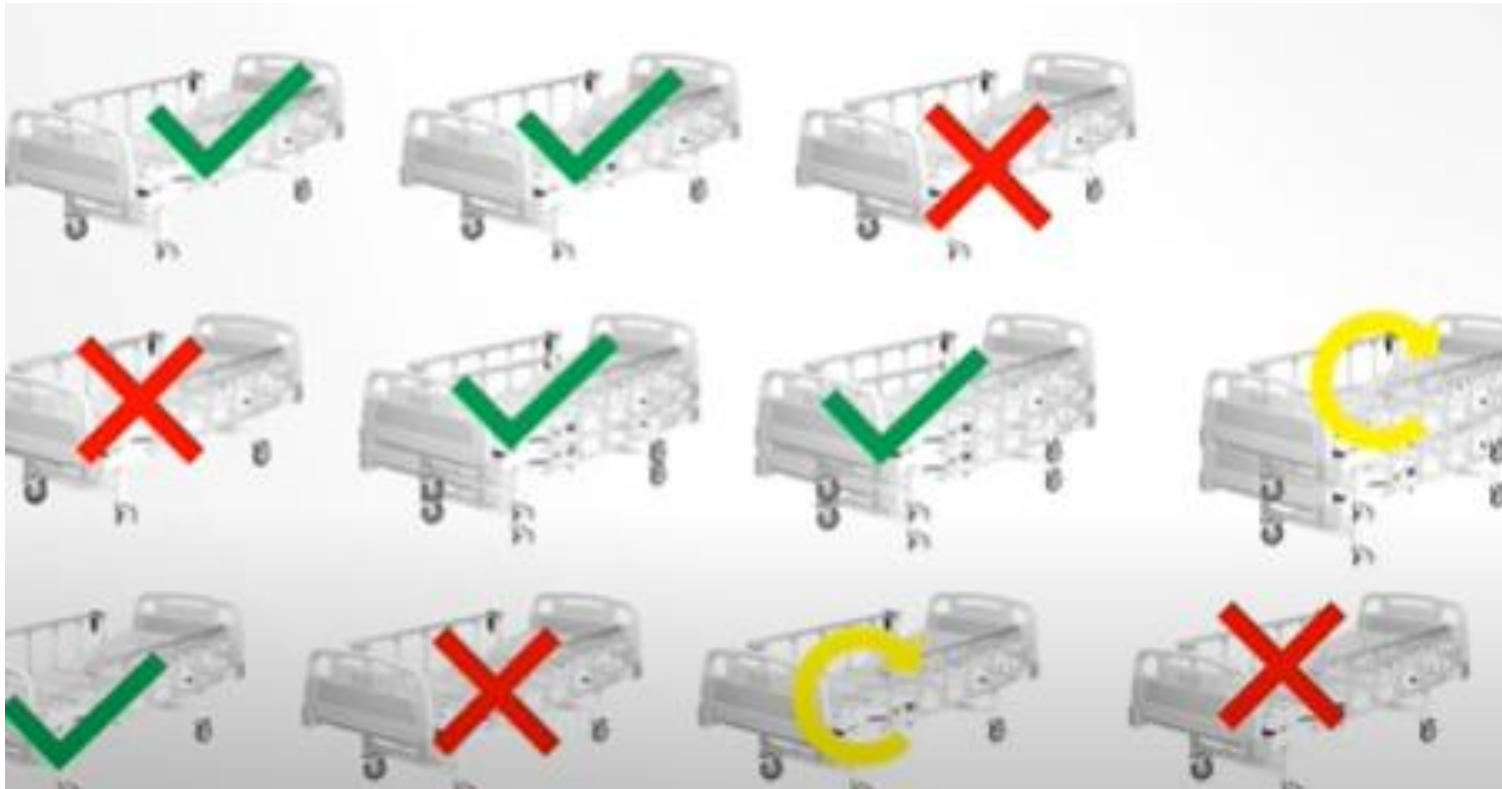
- This can be sometimes in the same fact table (Null for the non interesting grains and a value with the interesting ones)., or this also be moved into different tables based on the volumes and the design for the reporting purposes.
- In this type of Fact tables, we can incorporate any information that describes activity over a period like daily, weekly, monthly, etc.
- Just as transaction fact tables have many dimensions, the snapshot usually has fewer dimensions which lead it not to have the lower grain of data.
- This kind of Fact data will help in answering the questions like, How many days does a customer carry a positive account balance in one month? OR What is the average balance of a specific type of customer each month? etc.
- Two use cases for using **Periodic Snapshot Fact Table.**

Types of Fact Tables

❖ Periodic Snapshot Fact Tables.

• Two use cases:

- ❖ When we need to avoid many calculations, e.g., what described in the last slide about the different granularities.
- ❖ When need to follow up the STATUSES of very essential object. E.g., Bed Statuses in hospital every half an hour, every two hours,



Types of Fact Tables

❖ Periodic Snapshot Fact Tables.

Periodic Snapshot Fact Table:

- ❖ Stores current state of an essential object in the DW at a regular interval.
- ❖ Time Interval – Day, Week, Month or Even Few Hours or portion of an hour.
- ❖ Best Case – Analyze business performance at fixed interval.
- ❖ Common Examples: Bed Status in hospital, Stock Level at each node in few hours.
- ❖ Good to work with not Fully Additive Measures.

Types of Fact Tables

❖ Cumulative Snapshot Periodic Fact Tables

- ❖ A fact table that tracks the **lifecycle of a process** from start to finish.
- ❖ It stores multiple milestone dates in **one row**, updated as the process progresses.

❖ Characteristics

- One row per business process instance (e.g., one order).
- Updated over time.
- Contains multiple date foreign keys.
- Used to measure process performance.

Types of Fact Tables

❖ Cumulative Snapshot Periodic Fact Tables

❖ Example (Order Process)

Order ID	Order Date	Shipped Date	Delivered Date	Days to Ship	Days to Deliver
----------	------------	--------------	----------------	--------------	-----------------

As order progresses:

1. Row is updated with new milestone dates.
2. Durations are calculated.

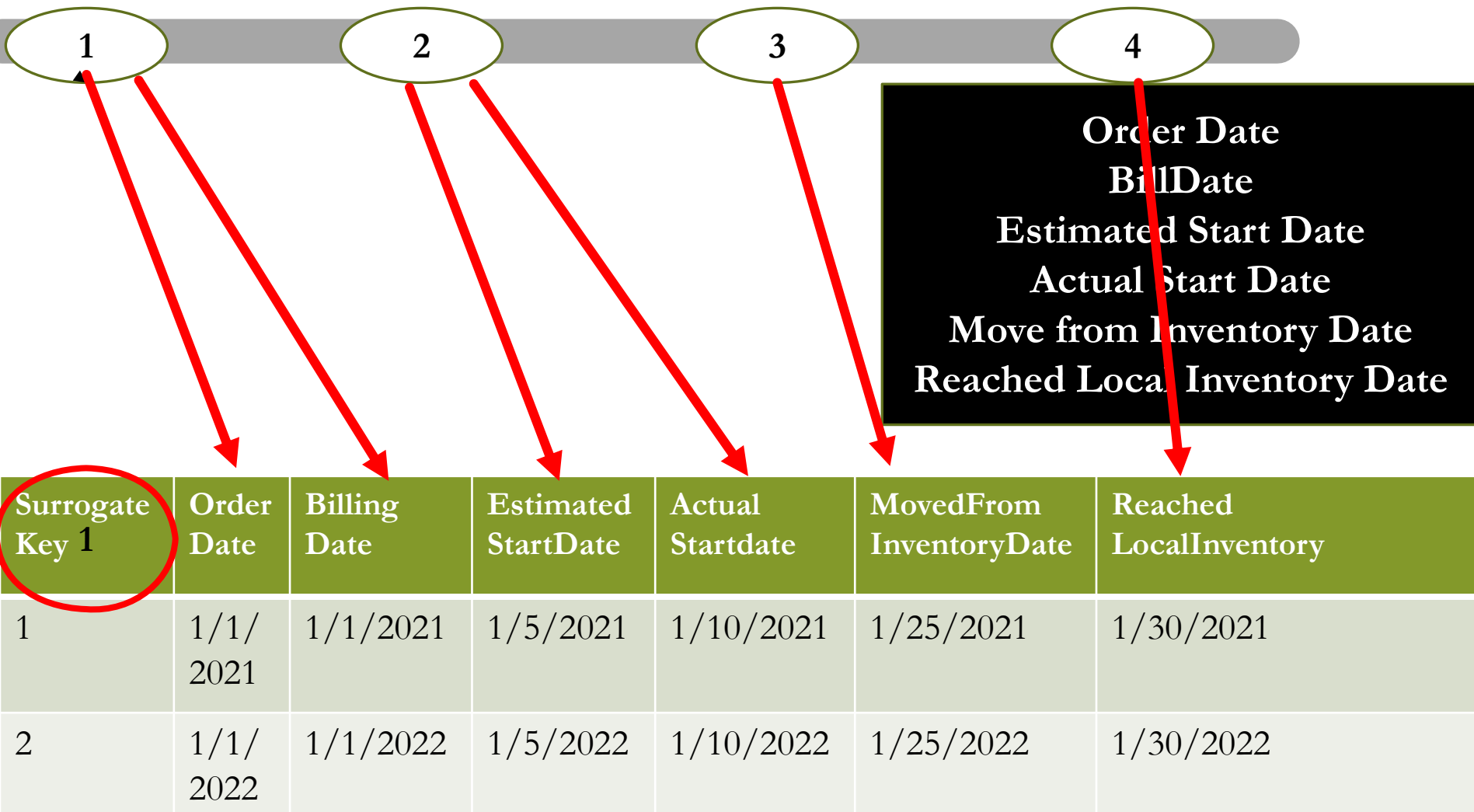
- **Business Use Cases**

- Order fulfillment tracking
- Shipment lifecycle
- Insurance claims process
- Manufacturing workflow stages

Types of Fact Tables

❖ **Cumulative Snapshot Periodic Fact Tables:**

❖ **Examples: Purchasing items, Materials, ...**



Types of Fact Tables

❖ Periodic VS Cumulative Snapshot Periodic Fact Tables

Feature	Periodic Snapshot	Cumulative Snapshot
Focus	State at regular intervals	Lifecycle of a process
Row Meaning	One row per time period	One row per process
Updates	Insert-only	Insert + update
Time Handling	Single date key	Multiple milestone dates
Table Density	Dense	Sparse
Example	Daily inventory	Order lifecycle